

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

```
#include <stdio.h>
```

```
int findDuplicate(int a[], int n) {
```

```
    int slow = a[0];
```

```
    int fast = a[0];
```

```
    do {
```

```
        slow = a[slow];
```

```
        fast = a[a[fast]];
```

```
    } while (slow != fast);
```

```
    slow = a[0];
```

```
    while (slow != fast) {
```

```
        slow = a[slow];
```

```
        fast = a[fast];
```

```
    }
```

```
    return slow;
```

```
}
```

```
int main() {
```

```
    int n;
```

```
    scanf("%d", &n);
```

```
    int a[n];
```

```
    for (int i = 0; i < n; i++) {
```

```
        scanf("%d", &a[i]);
```

```

}

printf("%d\n", findDuplicate(a, n));

return 0;
}

```

Input	Expected	Got	
	11 10 9 7 6 5 1 2 3 8 4 7	7	7
	5 1 2 3 4 4	4	4
	5 1 1 2 3 4	1	1

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For example:

Input	Result
5 1 1 2 3 4	1

```
#include <stdio.h>
```

```

int main() {
    int n;

    scanf("%d", &n);

    int a[n];

    int count[n + 1];

    for (int i = 0; i <= n; i++) {
        count[i] = 0;
    }
}

```

```

for (int i = 0; i < n; i++) {
    scanf("%d", &a[i]);
    count[a[i]]++;
}
for (int i = 1; i <= n; i++) {
    if (count[i] > 1) {
        printf("%d\n", i);
        break;
    }
}
return 0;
}

```

Input	Expected	Got	
	11 10 9 7 6 5 1 2 3 8 4 7	7	7
	5 1 2 3 4 4	4	4
	5 1 1 2 3 4	1	1

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

```
#include <stdio.h>
```

```
int main() {
```

```
    int T;
```

```
    scanf("%d", &T);
```

```
    while (T-->0) {
```

```
        int n1, n2;
```

```
        scanf("%d", &n1);
```

```
        int a[n1];
```

```
        for (int i = 0; i < n1; i++) {
```

```
            scanf("%d", &a[i]);
```

```
        }
```

```
        scanf("%d", &n2);
```

```
        int b[n2];
```

```
        for (int i = 0; i < n2; i++) {
```

```
            scanf("%d", &b[i]);
```

```
        }
```

```
        int i = 0, j = 0;
```

```

int first = 1;
while (i < n1 && j < n2) {
    if (a[i] == b[j]) {
        if (!first)
            printf(" ");
        printf("%d", a[i]);
        first = 0;
        i++;
        j++;
    }
    else if (a[i] < b[j]) {
        i++;
    }
    else {
        j++;
    }
}
printf("\n");
}
return 0;
}

```

Input	Expected	Got	
	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57
	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

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Example

Input:

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6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

```
#include <stdio.h>
```

```
int main() {
```

```
    int T;
```

```
    scanf("%d", &T);
```

```
    while (T-->0) {
```

```
        int n1, n2;
```

```
        scanf("%d", &n1);
```

```
        int a[n1];
```

```
for (int i = 0; i < n1; i++) {
    scanf("%d", &a[i]);
}
scanf("%d", &n2);
int b[n2];
for (int i = 0; i < n2; i++) {
    scanf("%d", &b[i]);
}
int i = 0, j = 0;
int first = 1;
while (i < n1 && j < n2) {
    if (a[i] == b[j]) {
        if (!first)
            printf(" ");
        printf("%d", a[i]);
        first = 0;
        i++;
        j++;
    }
    else if (a[i] < b[j]) {
        i++;
    }
    else {
        j++;
    }
}
printf("\n");
}
return 0;
}
```

Input	Expected	Got	
	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57
	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that $A[j] - A[i] = k$, $i \neq j$.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as $5 - 1 = 4$

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

```
#include <stdio.h>
```

```
int main() {
```

```
    int n, k;
```

```
    scanf("%d", &n);
```

```
    int a[n];
```

```
    for (int i = 0; i < n; i++) {
```

```
        scanf("%d", &a[i]);
```

```

}
scanf("%d", &k);
int i = 0;
int j = 1;
while (i < n && j < n) {
    if (i == j) {
        j++;
    }
    else {
        int diff = a[j] - a[i];
        if (diff == k) {
            printf("1\n");
            return 0;
        }
        else if (diff < k) {
            j++;
        }
        else {
            i++;
        }
    }
}
printf("0\n");
return 0;
}

```

Input	Expected	Got	
	3 1 3 5 4	1	1
	10 1 4 6 8 12 14 15 20 21 25 1	1	1

Input	Expected	Got	
	10 1 2 3 5 11 14 16 24 28 29 0	0	0
	10 0 2 3 7 13 14 15 20 24 25 10	1	1

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that $A[j] - A[i] = k$, $i \neq j$.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as $5 - 1 = 4$

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
int has_pair_with_diff(int A[], int n, int k) {
```

```
    int i = 0;
```

```
    int j = 1;
```

```
    while (i < n && j < n) {
```

```
        int diff = A[j] - A[i];
```

```
        if (i != j && diff == k) {
```

```

        return 1;
    } else if (diff < k) {
        j++;
    } else {
        i++;
        if (i == j) {
            j++;
        }
    }
}
return 0;
}

int main() {
    int n;
    if (scanf("%d", &n) != 1) return 0;
    int *A = (int *)malloc(n * sizeof(int));
    if (A == NULL) return 1;
    for (int i = 0; i < n; i++) {
        if (scanf("%d", &A[i]) != 1) return 0;
    }
    int k;
    if (scanf("%d", &k) != 1) return 0;
    int result = has_pair_with_diff(A, n, k);
    printf("%d\n", result);
    free(A);
    return 0;
}

```

Input	Expected	Got	
	3 1 3 5 4	1	1

Input	Expected	Got	
	10 1 4 6 8 12 14 15 20 21 25 1	1	1
	10 1 2 3 5 11 14 16 24 28 29 0	0	0
	10 0 2 3 7 13 14 15 20 24 25 10	1	1